Continuous Assessment-I

Computer Science and Engineering

Degree: B. E / B. Tech Semester: III Year: II

Course code & Title: CS23311 Computer Architecture

Date: 10.09.2025 QP CODE:071101

[Common to: Computer Science and Engineering]

[Regulations: 2023]

Time: 02.00 hour Max. Marks: 75

Answer All Questions

Part A: 10 x 2 = 20 Marks

1	Define CISC architecture.	[CO1]	[k1]
2	Why is RISC architecture preferred for embedded systems?	[CO1]	[k2]
3	Name any two addressing modes in ARM architecture.	[CO1]	[k1]
4	What is the purpose of logical operations in instruction execution?	[CO1]	[k1]
5	List any two functions of an ALU.	[CO2]	[k1]
6	Why is Booth's algorithm used in multiplication?	[CO2]	[k2]
7	What is the significance of normalization in floating-point representation?	[CO2]	[k1]
8	List any two differences between restoring and non-restoring division.	[CO2]	[k2]
9	Name the five stages in a MIPS pipeline.	[CO3]	[k1]
10	Why does pipelining improve processor performance?	[CO3]	[k2]

Part B: 5x11=55 Marks

11	a.	Compare RISC and CISC architectures with suitable examples. Include a labelled diagram of each.	[CO1]	[k2]
		[OR]		
	b.	Analyse the classic CPU performance equation and identify how each factor impacts execution time. Include a flow diagram.	[CO1]	[k2]
12	a.	Apply the concept of addressing modes to design an instruction set for a simple embedded application. Show examples of each mode in a table.	[CO1]	[k3]

[OR]

b.	Using ARM architecture as a case study, construct three sample instructions for decision making and illustrate their execution flow.	[CO1]	[k3]

a.	Apply the sequential multiplication algorithm to multiply two 4-bit numbers. Show all intermediate steps and draw the hardware flow diagram.	[CO2]	[k3]

[OR]

b.	Design an ALU section for performing addition, subtraction, and multiplication. Include a labelled logic block diagram	[CO2]	[k3]

a.	Compare the restoring and non-restoring division algorithms for signed numbers. Show step-by-step working for a sample input and compare with a diagram.	[CO2]	[k4]

[OR]

b.	Analyse floating-point multiplication for two IEEE 754 numbers and show the normalized result with a flow diagram.	[CO2]	[k4]

a.	Apply the MIPS single datapath to execute a load instruction. Draw the datapath diagram and highlight active components.	[CO3]	[k3]

[OR]

b.	Develop a pipelined datapath diagram for MIPS and identify how hazards are resolved in the pipeline stages.	[CO3]	[k3]